LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

Ch 5 Test (Set B): Life Processes

Chapter / Topic: Life Processes

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Draw neat, labelled diagrams wherever required.

SECTION A

6 marks

Choose the correct option.

Q1. Which of the following shows holozoic nutrition? **[1]**

(a) Cuscuta (b) Amoeba
(c) Mushroom (d) Green plants

Q2. The finger-like projections on the inner lining of the small intestine are called: **[1]**

(a) alveoli (b) nephrons
(c) villi (d) stomata

Q3. Which blood vessel carries deoxygenated blood from the heart to the lungs? **[1]**

(a) Pulmonary vein (b) Aorta
(c) Vena cava (d) Pulmonary artery

Q4. The opening and closing of stomata is caused by: **[1]**

(a) movement of the epidermis
(b) pressure in the xylem
(c) flow of food in the phloem
(d) a change in the water content of the guard cells

Q5. Which part of the nephron filters the blood? **[1]**

(a) Collecting duct (b) Glomerulus
(c) Ureter (d) Urethra

Q6. **Assertion (A):** Transport of materials in plants is slower than in animals.
Reason (R): Plants do not move, and many of their tissues are made of dead cells, so their energy needs are low. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. Why do aquatic organisms breathe much faster than terrestrial organisms? **[2]**

OR

Why do herbivores have a longer small intestine than carnivores?

Q8. What is the role of (a) the hairs and mucus in the nasal passage, and (b) the rings of cartilage in the throat (trachea)? **[2]**

SECTION C

6 marks

Q9. Answer the following about the human heart: **[3]**

(i) How many chambers does it have? **[1]**

(ii) Which chamber pumps oxygenated blood to the rest of the body? **[1]**

(iii) What is the function of the valves in the heart? **[1]**

Q10. Name the enzyme that digests each of the following, and say where in the alimentary canal it acts: (i) starch, (ii) proteins, (iii) fats. **[3]**

OR

How do plants get rid of their waste products? Give three ways.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) How are oxygen and carbon dioxide transported in human beings? **[2]**

(ii) Explain how we breathe in and breathe out. **[3]**

OR

Give the function of each of the following: (i) haemoglobin, (ii) bile, (iii) guard cells, (iv) glomerulus, (v) villi.

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. A potted plant was kept in a dark room for three days. One of its leaves was then partly covered with strips of black paper on both sides, and the plant was kept in sunlight for a few hours. The leaf was then boiled in alcohol to remove chlorophyll and tested with iodine solution. Only the uncovered parts of the leaf turned blue-black. **[4]**

(i) Why was the plant kept in the dark for three days? **[1]**

(ii) What does the blue-black colour with iodine show? **[1]**

(iii) What can you conclude from this experiment? Why did the covered parts not turn blue-black? **[2]**

~ All the best! ~